Review plan · for the lecturer

Teaching embedded and IoT systems at university: unit check

IoT · University · Unit check

Time	Materials
40 min	Projector or board, and the lecture slides.

Learning goals

- Sampling and the ADC
- Real-time and power
- Protocols
- Capstone design

Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

Discussion

- Which misconception will you target first?
- Which lab will you add?
- How will you change capstone assessment?

Exit ticket

- One change to my IoT course this term, and how I will know it worked.

Try it in your next lecture
Bring your capstone lab map and rubric to your next department meeting for review.

Quiz sheet · for students

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Name _____ Group _____ Date _____

1 **A signal has components up to 1 kHz. The minimum sampling rate is just above**

◯ 500 Hz ◯ 2 kHz ◯ 1 kHz

2 **Find the alias.**

An 80 Hz tone sampled at 100 Hz appears at _____ Hz.

Word bank: 180 · 20 · 80

3 **A digital filter after sampling can remove aliasing.**

◯ True ◯ False

4 **Active 40 mA for 1 s in 40, sleep ≈ 0. Average current:**

◯ 40 mA ◯ 1 mA ◯ 0.1 mA

5 **Which are hard real-time? Choose all that apply.**

☐ Anti-lock braking ☐ Airbag trigger ☐ Music streaming

6 **Match each MQTT QoS to its guarantee.**

QoS 0	• • • •	Exactly once
QoS 1	• • • •	At most once
QoS 2	• • • •	At least once

7 **Soil sensors over 5 km, tiny data, no power. Best link:**

◯ Wi-Fi ◯ LoRaWAN ◯ BLE

8 **Check the schedule.**

Tasks (C = 2, T = 10) and (C = 3, T = 15) give U = _____ %.

Word bank: 50 · 40 · 20

9 **Put the capstone preparation in order.**

☐ Proposal

☐ Scaffolded labs

☐ Final demo and defence

☐ Midpoint demo

10 **Best evidence of each student's contribution in a team:**

◯ Peer assessment plus logs ◯ Attendance ◯ The final demo

Answers · for the lecturer

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1. **2 kHz**
   fs > 2B = 2 kHz.

2. **An 80 Hz tone sampled at 100 Hz appears at 20 Hz.**
   |80 − 100| = 20.

3. **False**
   Aliases are indistinguishable from real low frequencies after sampling.

4. **1 mA**
   40 × 1/40 = 1 mA.

5. **Airbag trigger; Anti-lock braking**
   A late answer is a failure for both.

6. **QoS 0 — At most once**
   QoS 1 — At least once
   QoS 2 — Exactly once
   Higher guarantees cost more messages.

7. **LoRaWAN**
   Kilometres at low power and low rate.

8. **Tasks (C = 2, T = 10) and (C = 3, T = 15) give U = 40 %.**
   0.2 + 0.2 = 0.4.

9. **Top to bottom: 2, 1, 4, 3**
   Skills first, then the open problem in stages.

10. **Peer assessment plus logs**
    Several independent records are fairer.